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SOFTWARE DEVELOPMENT GROUP PROJECT

5COSCO021C

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Declaration page

I hereby declare that this project report and all the artifacts associated is my original work, completed as part of my academic requirements,and has not been previously submitted for any other academic program.

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Abstract + Keywords

A thorough three-chapter framework for system development is presented in this research.Where chapter 4 the system requirements specification (SRS) guide covers a Stakeholder analysis, Requirement elicitation techniques and methods, and a Use case diagram with prioritized functional and non-functional requirements. Whereas chapter 5 examines Social, Legal, Ethical and Professional issues (SLEP),with a focus on ethical clearance and mitigation techniques for the SLEP issues. Lastly, chapter 6's primary focus is System Architecture & design , which also contain thorough design components, illustrations, and a summary of the system architecture.This integrated approach ensures a thorough understanding of stakeholder need, ethical considerations, and a clear roadmap for system adoption

Acknowledgement

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Abbreviations Table

Abbreviation	**Explanation**
SRS	System Requirements Specification
UML	Unified Modeling Language

DFD	Data Flow Diagram

Table 01: Abbreviations Table

Chapter 4:System Requirements Specification (SRS)

4.1 Chapter Overview

This Chapter begins with a comprehensive stakeholder analysis using the onion model and detailed descriptions of each stakeholder group. It then explores the selection of appropriate requirement elicitation techniques and presents a critical analysis of the gathered insights. System functionalities are illustrated through use case diagrams and descriptions, providing a clear view of system interactions. Functional requirements are prioritized, while non-functional requirements highlight essential system constraints. A chapter summary consolidates these findings to support the next phases of the project.

4.2 Stakeholder Analysis

4.2.1 Onion Model

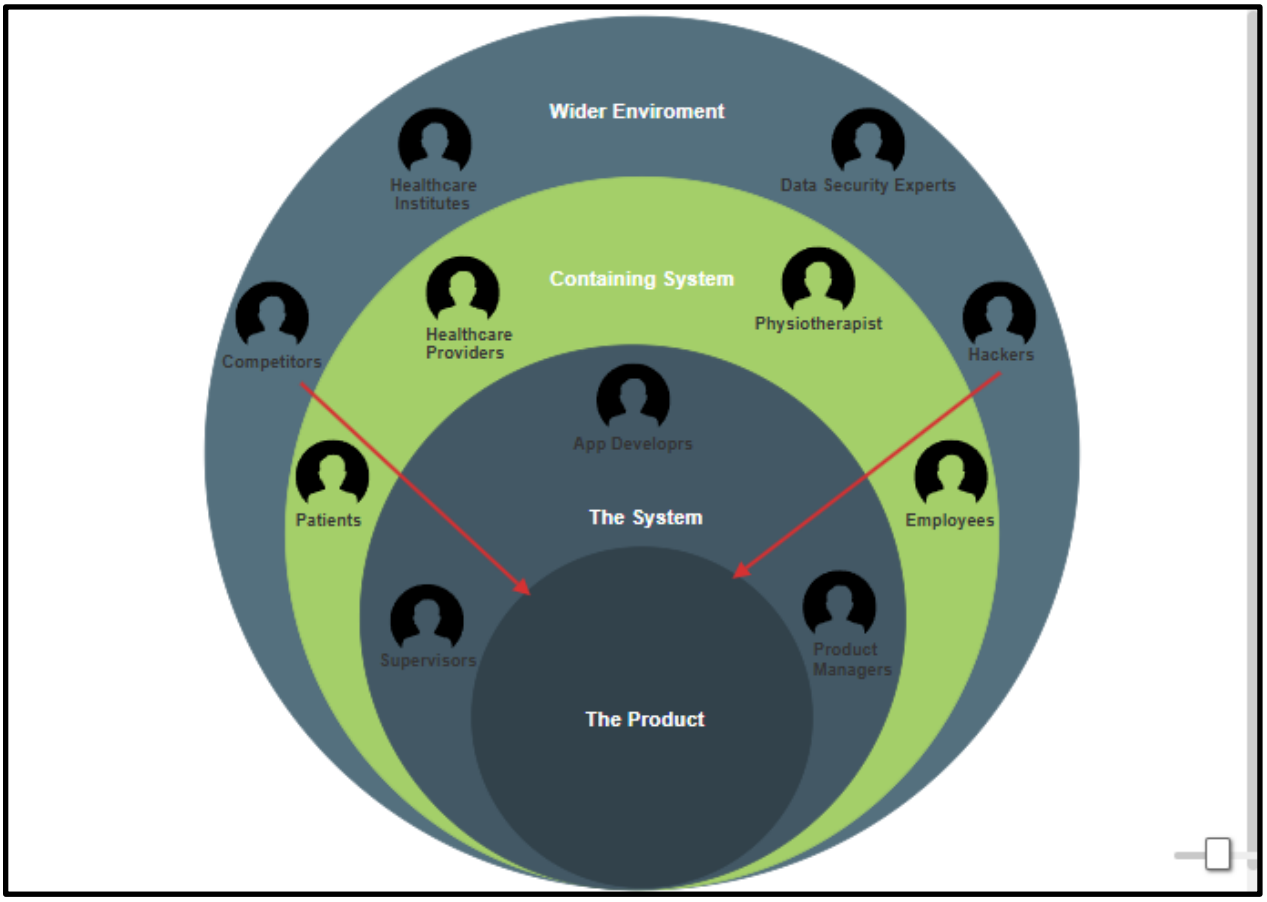


Figure 01: Onion Model

4.2.2 Stakeholder Descriptions

Stakeholder	Viewpoint
Functional Beneficiary	
Patients	Make advantage of easier access to physiotherapy treatments, such as personalized treatment plans and remote consultation.
Physiotherapists	Make use of the app to offer remote assistance, monitor patients progress, and broaden their reach.
Financial Beneficiary	
Owners	They can earn an income for themselves by charging for consultations and subscriptions and also from in-app revenues.
Healthcare providers	Since the services they provide are done virtually the operational cost such as equipment maintenance cost , luxury services cost can be reduced
Social Beneficiary	
Athletes	Offering rehabilitation resources,customized training regimens which helps them to speedup their recovery and increase performance.
General Users	Encourage proactive self-care health awareness by providing simple workouts to improve their overall quality of life.

Operational Beneficiary	
Healthcare Facilities	Provide services like virtual consultations, exercise tracking by integrating it into digital platforms that simplify patient care.
Employers	Increase employee satisfaction by integrating the app into workplace health programs to promote worker well-being.
Negative Stakeholders	
Traditional Clinic	A greater reliance on online physiotherapy could result in fewer in-person clinic visits, which would affect patient engagement on-site treatment revenue.
Competitors	Innovative physio apps with better user experience can be a threat to them and chances of losing their users.
Regulatory	
Health Authorities	Verify that the app complies with local or international data protection law like HIPAA.
Telemedicine Regulation	Provide rules and procedures to guarantee the safe and efficient delivery of remote healthcare services.
Experts	
Physiotherapy Specialist	For user effectiveness and engagement, incorporated personalized injury treatments,

	interactive training guidance
Technical Developers	Make responsiveness, easy navigation, and minimal downtime a top priority when it comes to performance.
Neighboring Systems	
Healthcare Information System	Facilitate the safe sharing of patient data by enabling smooth connection with EHR Systems.
Government Authorities	Ensure that the sector is financially sound, works equitably and complies with the law.

Table 02 : Stakeholder Descriptions

4.3 Selection of Requirement Elicitation Techniques / Methods

4.3.1 Interviews

Interviews are a widely used technique for obtaining requirements through direct interaction with stakeholders. To make sure that our Physio Connect app fits its goals, this strategy was used to learn about the requirements,expectations and worries of the app. Below are some methods on how it was done.

- Identifying Stakeholders
    - Making sure that every relevant stakeholder who has an impact on or influences on physiotherapy is involved from the beginning. By identifying these stakeholders,take their knowledge,to meet their requirements, and handle their issues. Which accelerates the app's success and acceptance by encouraging

cooperation, establishing trust, and guaranteeing that its functionality and design meet up to the expectations of our target consumers.

- Planning interviews
    - Organizing interviews guarantees effective, targeted, and fruitful engagement with our stakeholders. Where our team established expectations, clearly outlined goals, and developed an organized framework to collect insightful data by planning ahead and scheduling.

- Conducting One-on-One Interviews
    - Having targeted conversations to learn about requirements,goals, and difficulties. of our stakeholders who are,
        - Patients: Determine their preferences for convenient app use and virtual consultations.
        - Physiotherapists: Recognize the workflow requirements for efficient service.

- Group Interviews
    - Encouraging a cooperative atmosphere where stakeholders can express their distinct points of view,promoting the development of a variety of opinions of them like,
        - Patients: To increase engagement, talk about common problems and desired features.

4.3.2 Surveys and Questionnaires

Questionnaires and surveys are instruments for collecting both qualitative and quantitative information from a specific audience. They include structured or semi-structured questions that assist identifying user preferences, behaviors, and needs.

- Defining Objectives:
    - Establishing clear goals and objectives to guarantee that the survey collects useful data that can be out to use. By concentrating on the requirements of various stakeholders, the survey aims to find user preferences, difficulties, and areas for innovation, which will help in improving the app's features and bringing them into line with the user expectations.

- Select Respondents:
    - To guarantee thorough and fair input, choosing a diverse set of stakeholders. The survey's inclusion of patients, physiotherapists, healthcare admins allows to gather a variety of viewpoints,guaranteeing that the app successfully satisfies the needs of all stakeholders and potential users.

- Design Questions:
    - Create focused, direct enquiries to collect relevant information. The survey will gather important input on user requirements, app usability and desired features using clear, straightforward questions. This will guarantee that the information collected right away affects the creation and improvement of the Physio Connect App.

- Distribute Surveys:
    - To make sure the survey effectively reaches a large audience, distribute it across a variety of approaches.throughout the use of social media, in-app notifications,

email, and specific healthcare groups, the survey will collect a variety of responses, guaranteeing thorough insights that helps in the growth of the app.

4.3.3 Use Cases Scenarios

Use case scenarios explain how users engage with a technology to accomplish particular objectives. They give an in-depth description of the actions a user conducts, emphasizing the capabilities of the system and pointing out any possible problem or area for development. Use case scenarios are useful for comprehending user needs and coordinating system design with particular applications.

4.3.4 Prototyping

Making an early prototype of a system or a product in order to see and test its functionality is known as prototyping. Prior to full scale development, it helps refine the design and gives stakeholders a chance to offer feedback. Depending on the needs and stage of the project, prototypes can be anything from low-fidelity sketches to high-fidelity interactive models. Below are some ways on how we implement the prototyping technique.

- Ui design - we built a prototype using figma which is added under the section 6.3.3 UI Design and mockup where it gives a rough idea on how our application will run and the features included in it.

4.3.5 Focus Groups

Focus groups are discussions with a small group of participants that are monitored in order to learn more about their expectations, experience and opinions. In addition to offering qualitative information on user requirements, preferences, and possible enhancements,

4.3.6 Document Analysis

Examining current documentation such as technical specifications, policies, or user manuals in order to gather relevant details about a system or procedure is known as document analysis. By analyzing current workflows, compliance needs and documentation gaps,

4.4 Discussion / Analysis of Results

4.4.1 Interview Analysis

We interviewed physiotherapists and athletes to understand their needs for Physio Connect App.where we discussed with physiotherapists their expectations for the app and what features they felt essential also we discussed with athletes on their rehabilitation coud be aided with a digital solution

4.4.2 Survey & Questionnaire Analysis

Question 01

Question 01 Analysis

Analysis is not available for this question.Since it was a collection of emails of respondents.

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Question 02

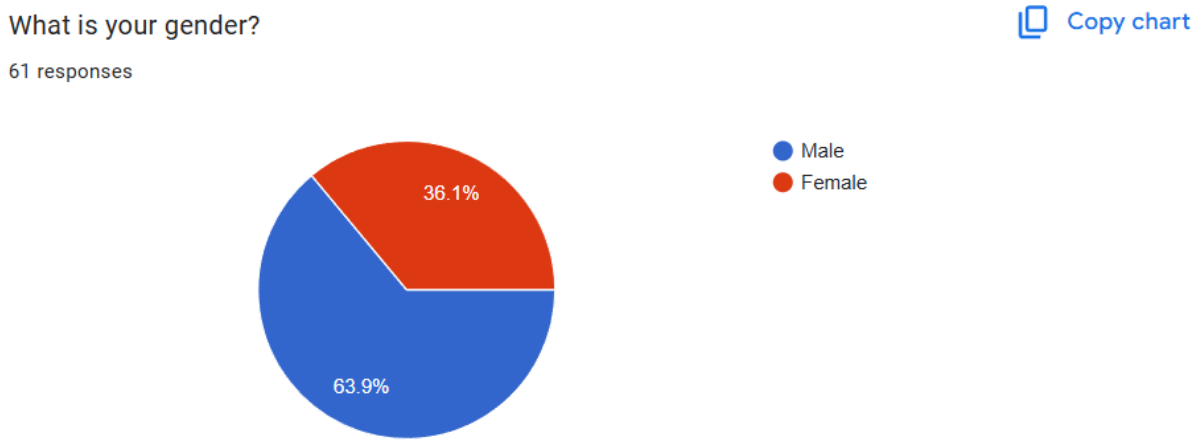


Figure 02 : Survey Q2

Question 02 Analysis

According to the response we got, the majority of boys answered the survey for the physio connect app.which is 63.9%out of the total response received.

Question 03

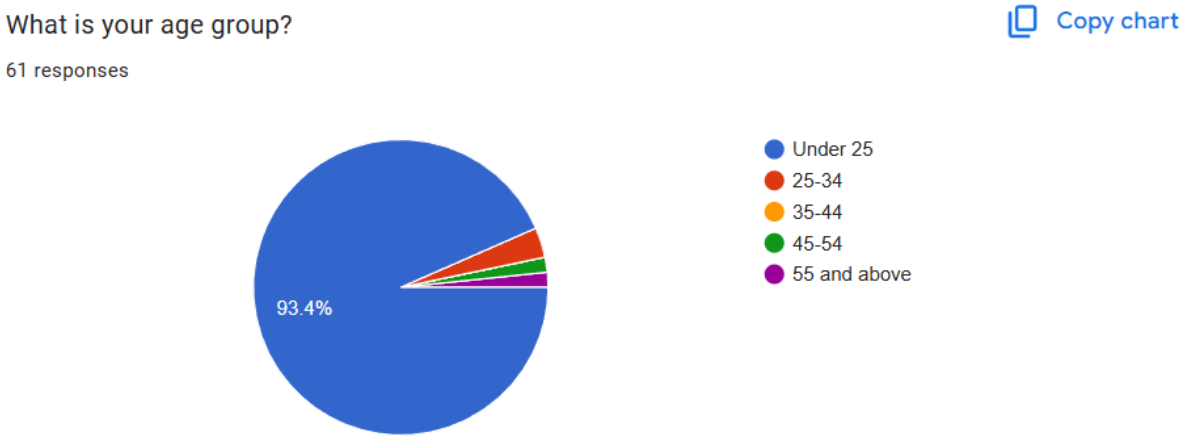


Figure 03 : Survey Q3

Question 03 Analysis

According to the response we got, the majority of the age group is people below 25 who will be using the physio connect app.

Question 04

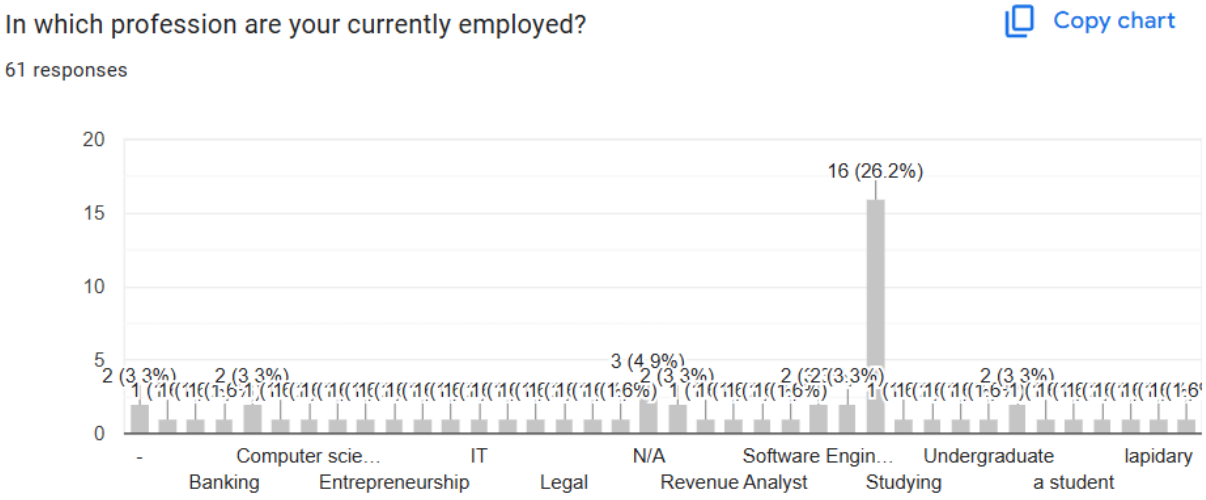


Figure 04: Survey Q4

Question 04 Analysis

From the following responses we have got the most of the responses from the students studying in universities, schools and colleges.

Question 05

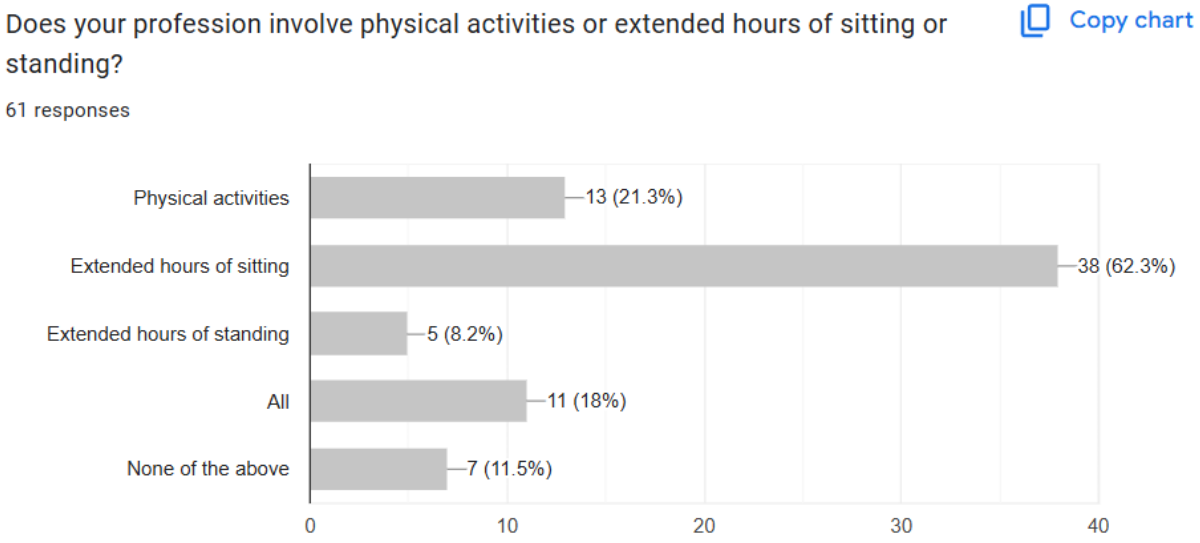


Figure 05 : Survey Q5

Question 05 Analysis

According to the responses we can come to a conclusion that most of the people who filled the survey have long sitting hours in their day to day life.

Question 06

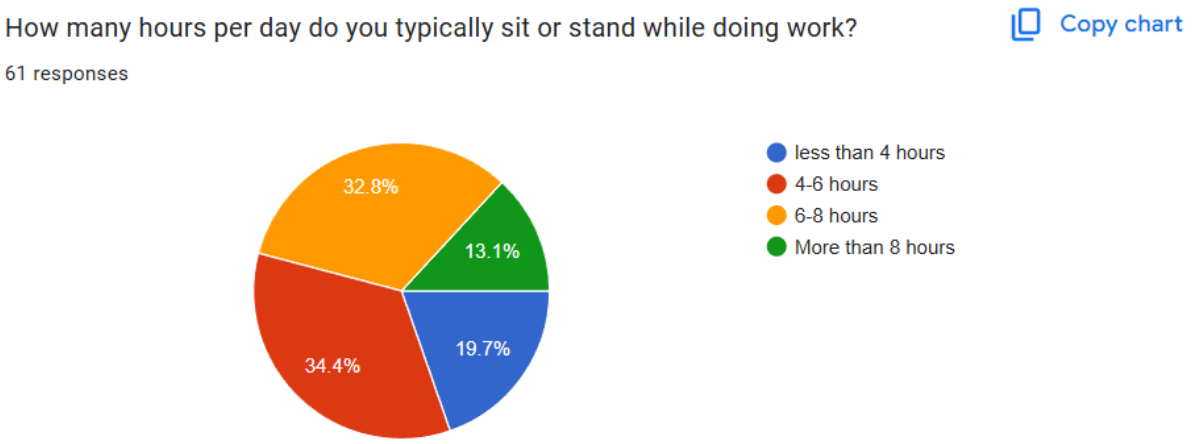


Figure 06 : Survey Q6

Question 06 Analysis

From the people with long sitting hours 13.1% of our respondents sit/stand for more than 8 hours per day, where 34.4% are with 4-6 hours a day.

Question 07

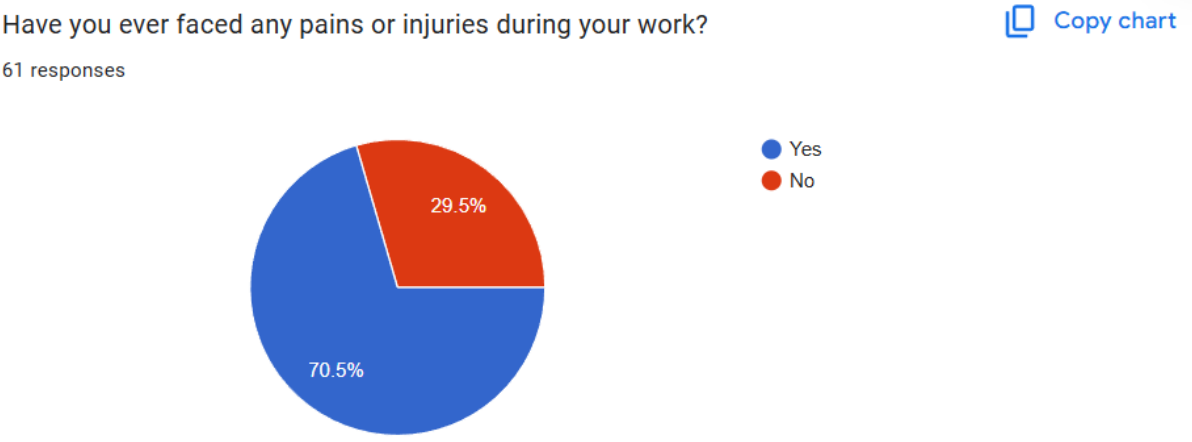


Figure 07 : Survey Q7

Question 07 Analysis

70.5% of the respondents have faced pains and injuries during work

Question 08

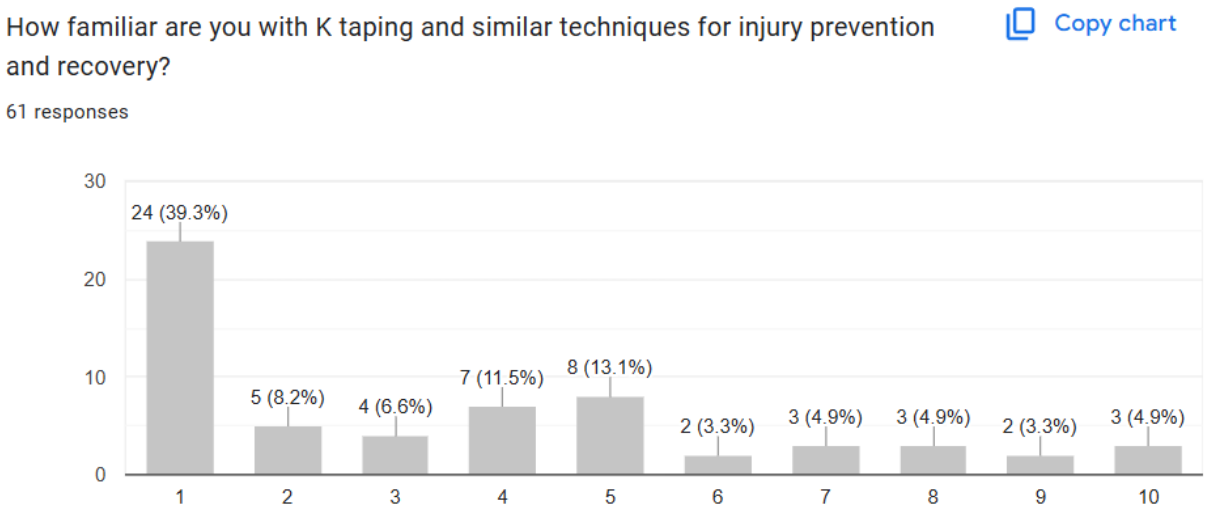


Figure 08 : Survey Q8

Question 08 Analysis

A few of the respondents has an idea on K taping and injury prevention and recovery techniques

Question 09

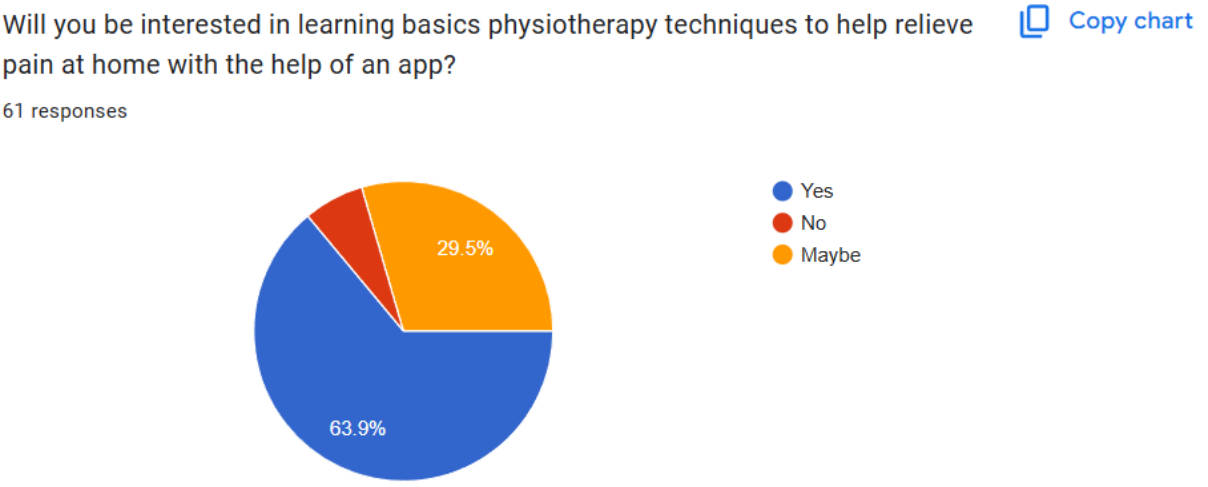


Figure 09 : Survey Q09

Question 09 Analysis

Most of the respondents are willing to learn physiotherapy techniques to ease life and help to relieve their pain at home.

Question 10

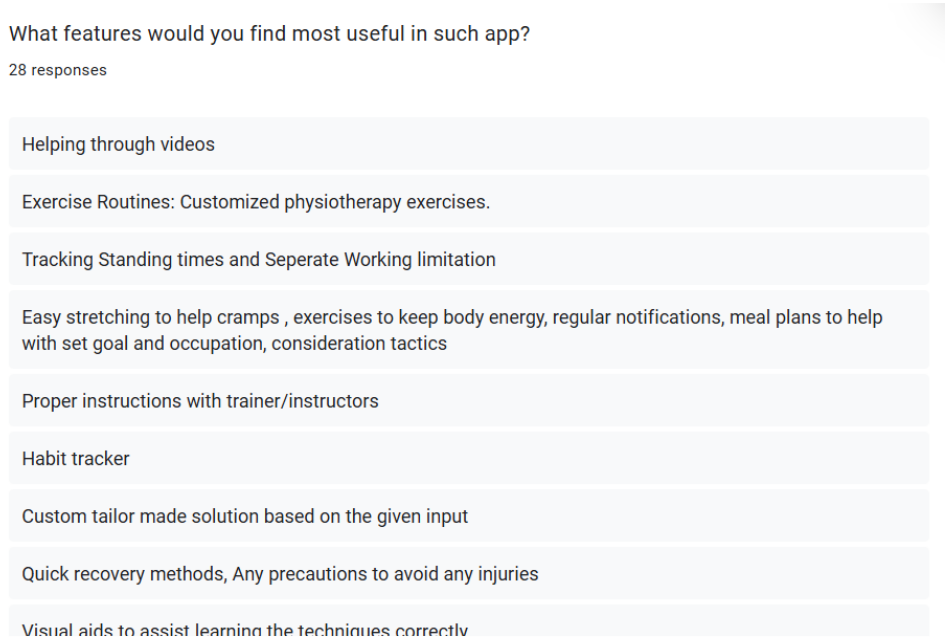


Figure 10 : Survey Q10 (A)

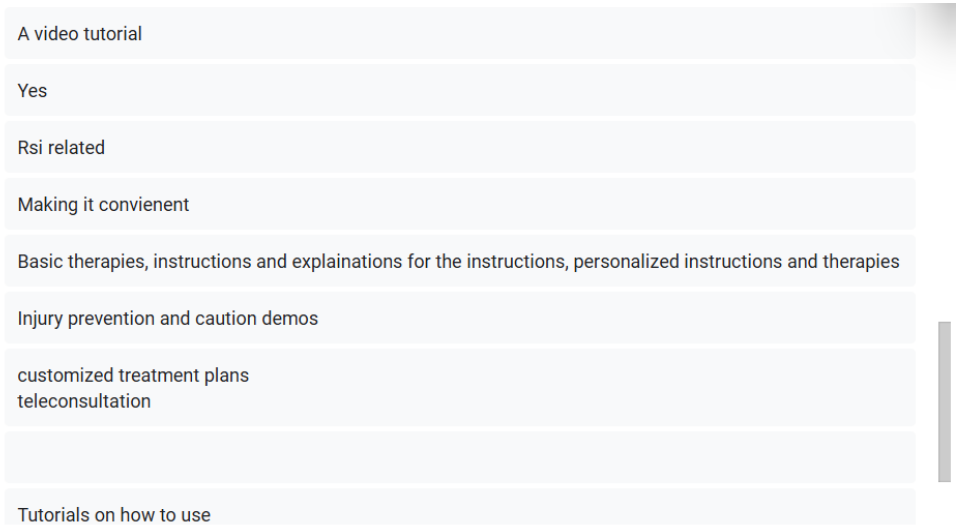


Figure 11 : Survey Q10 (B)

Question 10 Analysis

Here are some responses given by respondents about the most useful feature to be included in the app to make it more effective.

Question 11

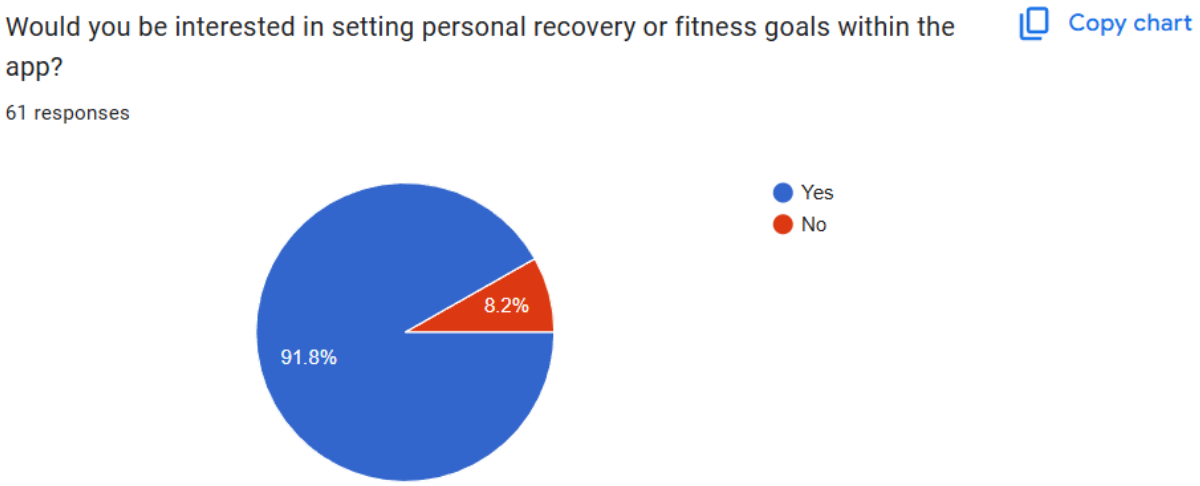


Figure 12 : Survey Q11

Question 11 Analysis

Majority of the respondents would strongly like to have personal recovery or fitness goals within our app.

4.5 Use Case Diagram

The following Use Case Diagram will provide the basic understanding of how users will interact with the PhysioConnect App.

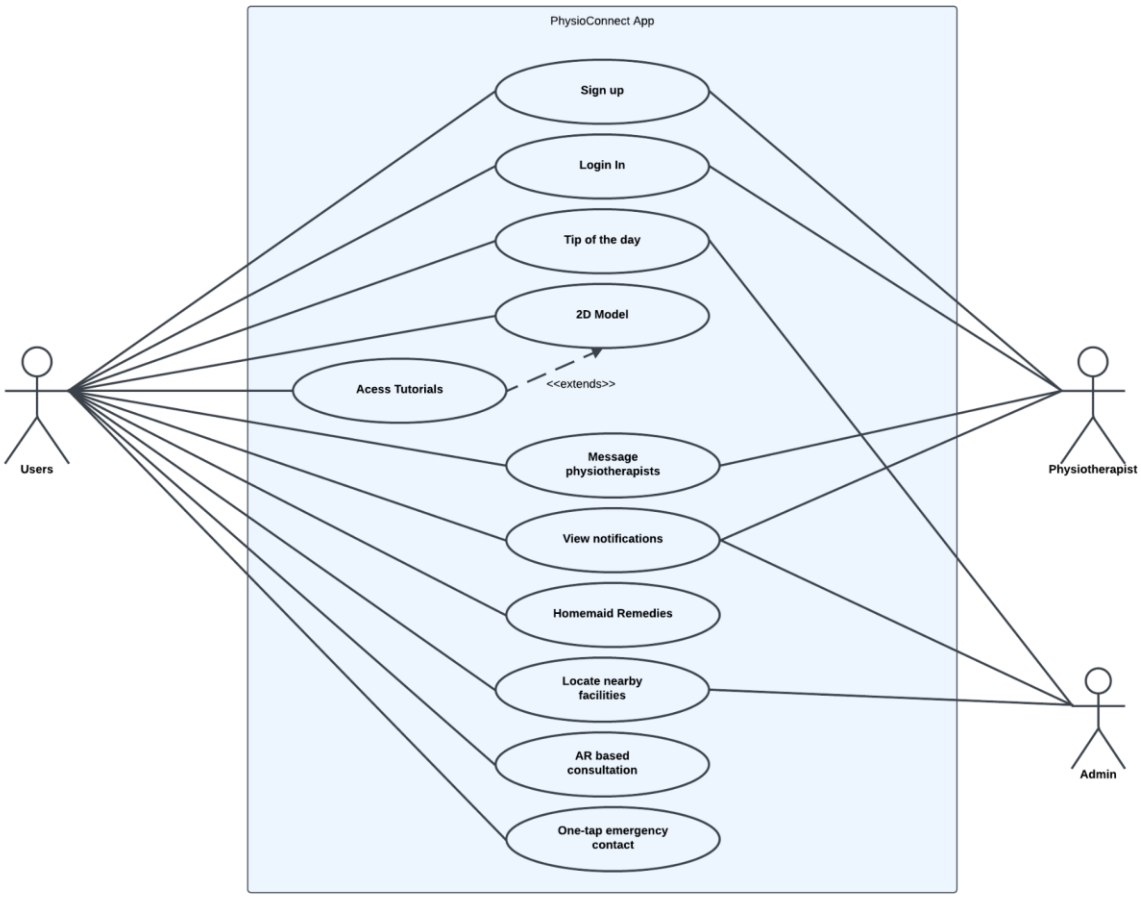


Figure13 : Use case Diagram

4.6 Use Case Description

4.6.1 Sign Up

Use Case Name	Sign up	
Use Case ID	UC-001	
Description	Users and Physiotherapists can register themselves and create an account for themselves.	
Priority	High	
Primary Actor	User, Physiotherapist	
Supporting Actors	None	
Pre-Conditions	The application has to be installed by the user /physiotherapist.	
Trigger	The user/physiotherapist chooses the “Sign Up” option upon opening the app..	
Main flow	Actors	System
	1. The user/physiotherapist enters a username and a strong password to create an account.	2. The system verifies the details entered 3. It creates a new account for the users/physiotherapist.
Exception flow	Actors	System

	1. If the information entered are incorrect	2. The user or the physiotherapist is prompted by the system to make changes.
Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	The users or the physiotherapists can login since they have an account that is registered.	

Table 03 : Use Case Description - Sign Up

4.6.2 Log In

Use Case Name	Log In	
Use Case ID	UC-002	
Description	The registered users/physiotherapists enter their login information to access the physio connect app.	
Priority	High	
Primary Actor	User, Physiotherapist	
Supporting Actors	None	
Pre-Conditions	The user/ physiotherapist should know their login credentials.	
Trigger	The user/physiotherapist chooses the ‘Login’ option.and enter their credentials.	
Main flow	Actors	System

	1. user/physiotherapist input their username and password.	2. The credentials are verified by the system. 3. The system provides access for the application if it is vaid.
Exception flow	Actors	System
	1. When the entered credentials aren't valid	2. The user/physiotherapist is prompted by the system to enter valid credentials.
Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	The user/physiotherapist is logged in and can use the app features.	

Table 04 : Use Case Description - Login

4.6.3 Tip of the Day

Use Case Name	Tip of the Day
Use Case ID	UC-003
Description	This is where physiotherapy related tips are given to user
Priority	Medium
Primary Actor	User
Supporting Actors	Admin

Pre-Conditions	The user should have logged in to the app	
Trigger	Click on the 'Tip of the day' section.	
Main flow	Actors	System
	1. User clicks on the tip of the day option.	2. Fetches data from the database an 3. Displays it on the screen
Exception flow	Actors	System
	None	None
Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	Users can access daily tips	

Table 05 : Use Case Description - Tip of the Day

4.6.4 2D Model

Use Case Name	2D Model
Use Case ID	UC-004
Description	This is where the user click on a specific area on the 2d model
Priority	High
Primary Actor	User
Supporting Actors	None

Pre-Conditions	The user must login to the application	
Trigger	Click on the area which is needed for treatment on the model	
Main flow	Actors	System
	1. The user clicks on the area he/she needs therapy	2. The system gives the required treatment plans and give tutorials
Exception flow	Actors	System
	None	None
Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	The user gets the needed tutorial and recovery treatment plans for the injury.	

Table 06 : Use Case Description - 2D Model

4.6.5 Message Physiotherapists

Use Case Name	Message Physiotherapists
Use Case ID	UC-005
Description	Users can message physiotherapists for advice or for any clarification.
Priority	Medium
Primary Actor	User

Supporting Actors	Physiotherapist, Admin	
Pre-Conditions	User and the physiotherapist should have logged in.	
Trigger	Click on the ‘message physiotherapist’ option.	
Main flow	Actors	System
	1. The user selects a specific physiotherapist. 2. The user inputs the message to be sent	3. The system sends the message to the selected physiotherapist
Exception flow	Actors	System
	1. If the physiotherapist is not available.	2. The system will informs the user about it.
Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	Gets a response from the physiotherapist for the question.	

Table 07 : Use Case Description - Message Physiotherapists

4.6.6 Access Tutorials

Use Case Name	Access Tutorials
Use Case ID	UC-006
Description	Once the user clicks on the 2D model at a specific area for

	treatment it gives tutorials related to it.	
Priority	High	
Primary Actor	User	
Supporting Actors	None	
Pre-Conditions	Have logged in to the app	
Trigger	Click on the area in the 2D model	
Main flow	Actors	System
	1. The user click on the 2D model on the area he/she is having pain	2. The system gives out tutorial and other treatment plans for the pain recovery
Exception flow	Actors	System
	None	None
Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	User receives tutorials related to his/her injury for recovery	

Table 08 : Use Case Description -Access Tutorials

4.6.7 View Notification

Use Case Name	View Notification
Use Case ID	UC-007

Description	The user/ physiotherapist receives notification from the software for related activities.	
Priority	High	
Primary Actor	User, Physiotherapist	
Supporting Actors	Admin	
Pre-Conditions	Must have enabled app notifications.	
Trigger	The user/physiotherapists select the notifications	
Main flow	Actors	System
	1. User receive notification from the system	2. System extract them from the database 3. Display them to the users
Exception flow	Actors	System
	None	None
Alternate flow	Actors	System
	none	None
Exclusions	None	
Post Conditions	User or Physiotherapist are update with schedule of consultation sessions	

Table 09 : Use Case Description - View Notification

4.6.8 Home Remedies

Use Case Name	Home Remedies	
Use Case ID	UC-008	
Description	This shows the user the remedies that he/she can take at an injury at home.	
Priority	High	
Primary Actor	User	
Supporting Actors	None	
Pre-Conditions	Must input the data about the injury to the system	
Trigger	Select the home remedy option	
Main flow	Actors	System
	1. User select home remedies option	2. The system gives out some remedies using ai and other references
Exception flow	Actors	System
	1. If the injury cannot be recovered using home remedies	2. The system will update the user
Alternate flow	Actors	System
	None	None
Exclusions	None	

Post Conditions	The remedies are provided to the user.

Table 10 : Use Case Description - Home Remedies

4.6.9 Nearby Facility Locator

Use Case Name	Nearby Facility Locator	
Use Case ID	UC-009	
Description	This allows users to find nearby health facility centers at any injury or treatment.	
Priority	High	
Primary Actor	User	
Supporting Actors	None	
Pre-Conditions	Must log in to the app.	
Trigger	Find nearby	
Main flow	Actors	System
	1. The user selects the nearby facility locator option	2. The system will access the user's location 3. Then it will give the nearby facilities in his/her area.
Exception flow	Actors	System
	1. If there are no any facility center nearby	2. The system will notify the user

Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	The user will view the nearby facility.	

Table 11 : Use Case Description - Nearby Facility Locator

4.6.10 AR Based Consultation

Use Case Name	AR Based Consultation	
Use Case ID	UC–010	
Description	Allows the user to consult physiotherapists through augmented reality.	
Priority	High	
Primary Actor	User	
Supporting Actors	Physiotherapist	
Pre-Conditions	Both users and physiotherapists must have AR compatible devices.	
Trigger	Select the ‘AR Based Consultation’ Option	
Main flow	Actors	System
	1. The user select AR consultation option	2. The system connects a physiotherapist

Exception flow	Actors	System
	1. If the service in not available	2. The system will notify the user about the unavailability
Alternate flow	Actors	System
	None	
Exclusions	None	
Post Conditions	The user will get consulted via AR	

Table 12 : Use Case Description - AR Based Consultation

4.6.11 One-tap Emergency Contact

Use Case Name	One-tap Emergency Contact
Use Case ID	UC-011
Description	Allows users to quickly contact any emergency service or any emergency contact saved.
Priority	High
Primary Actor	User
Supporting Actors	None
Pre-Conditions	Must have at least one number saved in the emergency contact list.
Trigger	The user select the One-Tap Emergency Contact button

Main flow	Actors	System
	1. The user select the emergency contact	2. The system dials the emergency contact
Exception flow	Actors	System
	1. If no number is saved in the emergency contact list	2. The system asks the user to enter a contact to the list.
Alternate flow	Actors	System
	None	None
Exclusions	None	
Post Conditions	The user's emergency contact is contacted.	

Table 13 : Use Case Description - One-tap Emergency Contact

4.7 Functional Requirements (with prioritization)

The following table will discuss the function requirements of this project, as well the priority level of each requirement.

Critical - The Requirements that are critically needed for successful completion

Desirable - The Requirements that can add value, but are not required immediately

Luxury - The Requirement that would add luxury to the system

Requirement List		Priority Level	Description
FR1	User Registration/ Login	Critical	This allows users to create accounts and securely login in,storing personal and health-related data.
FR	Customized Exercise Plan	Critical	This generates tailored exercise routines based on user injury details and recovery goals.
FR3	Progress Tracking	Critical	This allows users to monitor their rehabilitation journey with detailed insights.
FR4	Educational Resources	Critical	This provides articles,tutorials on injuries, recovery methods and preventive measures for the users.
FR5	2D Body Model	Critical	Through this interactive model it helps the user to locate injuries and provide region-specific advice and exercise.
FR6	Book Appointments	Desirable	This allows users to book appointments with physiotherapists.
FR7	Video Consultations	Desirable	This enables remote consultations with physiotherapists for assessments and

			treatment plans.
FR8	Electronic Personal Health Record	Desirable	This centralized users health data, including injury history, progress and consultation notes.
FR9	Messaging System	Desirable	This enables communication between users and physiotherapists for support.
FR10	Nearby Facilities Locator	Desirable	Through this it helps users to find nearby physiotherapy centers or medical facilities using geo-location tools.
FR11	Push Notification	Desirable	Through this it sends reminders to the users for appointments,exercise and daily health tips to enhance compliance.
FR12	Emergency Contact	Desirable	This is a one-tap access to emergency contacts for immediate assistance.
FR13	AI Chatbot for Advice	Luxury	This offers real-time physiotherapy tips and solutions via an AI powered chatbot.
FR14	Payment Integration	Luxury	Through this patients can pay for the packages directly within the app.
FR15	AR Implementation	Luxury	This uses augmented reality to guide users through exercises with real-time posture and movement.
FR16	Gamification	Luxury	This includes features like points and challenges to boost user engagement.

Table 14 : Functional Requirements

4.8 Non-Functional Requirements

The following table will discuss the non functional requirements of this project, as well the priority level of each requirement.

Critical - The Requirements that are critically needed for successful completion

Desirable - The Requirements that can add value, but are not required immediately

Luxury - The Requirement that would add luxury to the system

Requirement list		Priority Level	Description
NFR1	Response Time	Critical	This is where the app should respond to user actions in a short period of time to improve user experience.
NFR2	Scalability	Critical	This is where the app must support increasing user volume without performance issues.
NFR3	Data Encryption	Critical	During Storage user data must be encrypted.
NFR4	Authentication	Critical	This uses a safe authentication system to protect user accounts.
NFR5	Availability	Critical	This ensures 99.9% uptime with minimal disruptions during maintenance.
NFR6	Accessibility	Desirable	This Complies with standards to ensure usability for all,including disabled users.

NFR7	Backup & Recovery	Desirable	This performs daily backups and ensures recovery in case of system failure.
NFR8	Cross-Platform Compatibility	Desirable	This provides a consistent experience across a range of operating systems.
NFR9	User Training Resources	Luxury	This provides materials to help users utilize the app's features fully.
NFR10	Performance Monitoring	Luxury	This Implements monitoring tools to supervise user interaction and system performance.

Table 13 : Non-Functional Requirements

4.9 Chapter Summary

For any project to run smoothly, the system requirements must be evaluated.The ability to accurately define the system requirements is one of the main pillars of making our project a successful one. This chapter focuses on the stakeholders which was presented using a onion model followed by the descriptions and analysis, the choice of elicitation techniques used for the project, a discussion,functional and nonfunctional requirements with their respective priority levels, and an examination of the questionnaiaare's findings Additionally, the use of case diagrams and descriptions that outline our project's main ideas are implemented.

Chapter 5: Social, Legal, Ethical and Professional Issues

5.1 Chapter Overview

This Chapter explores our project's wider implications, highlighting its interactions with societal, legal, ethical and professional aspects. It draws attention to the obligations placed on us admins and other stakeholders to uphold professional conduct. Address any social effects, and guarantee adherence to legal and ethical norms. These factors are essential for creating trust, defending rights, and guaranteeing our project's sustainability.

5.2 Dataset Ethical Clearance

The datasets used in this project are for the identification process and information are taken from Kaggle (www.kaggle.com/), The Creative Commons Licence Organization has granted it a Public Domain license, indicating that the general public may access and use it for research purposes:

Patient-Related Dataset:

Information about patient physical activity tracking and chronic pain management.
https://www.kaggle.com/search?q=Chronic+Pain+and+Physical+Activity

General Healthcare Datasets

includes a broad range of health information that may be utilized to examine patterns in patient recovery.
https://www.kaggle.com/search?q=Health+and+Lifestyle+Data

Exercise and Rehabilitation Datasets

dataset of smartphone sensors used to identify human activities. It contains information that may be relevant to exercises used in physical therapy.
https://www.kaggle.com/search?q=Human+Activity+Recognition

5.3 SLEP Issues and Mitigation

When developing software projects, social, legal, ethical, and professional issues must be taken into consideration, the topics below will help you understand how, during the development of our Physio Connect app, social, legal, ethical, and professional issues were handled in accordance with the BCS code of conduct .

5.3.1 Social Issues

By providing virtual consultations and rehabilitation tools, the Physio Connect app aims to increase access to physiotherapy services,particularly in underserved or rural communities. However, its dependence on smart phone access and internet connectivity can put those in low-income neighborhoods or areas with weak digital infrastructure at a disadvantage. For people who are not technologically proficient, such as older people or those living in places with inadequate network coverage, this digital gap may limit their access to necessary physiotherapy services. Although efforts to increase accessibility in rural areas may encounter difficulties, the project will mainly focus on metropolitan areas and semi-urban areas where technology access is more common.

5.3.2 Legal Issues

By matching patients with certified physiotherapists, the Physio Connect app seeks to provide outstanding therapy. It must, however, manage complex healthcare regulations, such as making sure data protection rules (including HIPAA and GDPR) are followed and obtaining the appropriate licensing in various jurisdictions. Legal issues related to professional license, liability, and different regional healthcare regulations may come up during international consultations. Important legal issues to solve including obtaining informed permission and protecting patient privacy during virtual sessions.

5.3.3 Ethical Issues

In order to provide users with individualized care, the Physio Connect app collects sensitive health data, which raises ethical questions regarding data usage and privacy. Retaining trust involves making sure that user’s data is not taken advantage of or shared without informed agreement. Furthermore, it is ethically necessary to protect AI-driven recommendations against biases that can unintentionally give preference to particular groups or conditions over other individuals, which could result in unfair treatment outcomes.

5.3.4 Professional Issues

To provide high quality therapy, the Physio connect app must guarantee that all physiotherapists on the platform adhere to professional licensing and accreditation requirements. Verifying and upgrading credentials across multiple platforms is a major task, particularly when laws and requirements differ. Furthermore, physiotherapists must effectively manage patients expectations in a virtual setting while providing therapy within their scope of practice, which can make maintaining accountability for remote consultations difficult. This requires constant education and adherence to professional standards in order to maintain the platform's credibility.

5.4 Chapter Summary

The social, legal, ethical and professional difficulties that our project is likely to face are covered in this chapter. As the project is carried out these difficulties will become more noticeable. Our team determined and examined the relevant SLEP issues for our work in order to address this. In order to ensure that the initiative avoids adverse effects from outside sources and minimize any potential harm it may cause in return,Our team also specified steps to prevent these worries.

Chapter 6: System Architecture & Design

6.1 Chapter Overview

This chapter focuses on the foundational aspects of designing and organizing a software system's structure and behavior. It examines the system's architecture and design for our Physio Connect app, including its components, interface, and interactions. The goal of this chapter is to provide a detailed understanding of system design of our app, which involves defining the architecture, components, connections and the data flow with the system.

6.2 System Architecture Design

One of the most widely used architectural designs is layered architecture. The concept of layered architecture is to create horizontal levels by assembling sections or components with related functions. Below is the architectural design of our Physio connect app where the first layer is the Presentation layer. The second layer is called the Domain Logical layer which represents the core functionalities. And last is the Data Storage layer which represents the data stored.

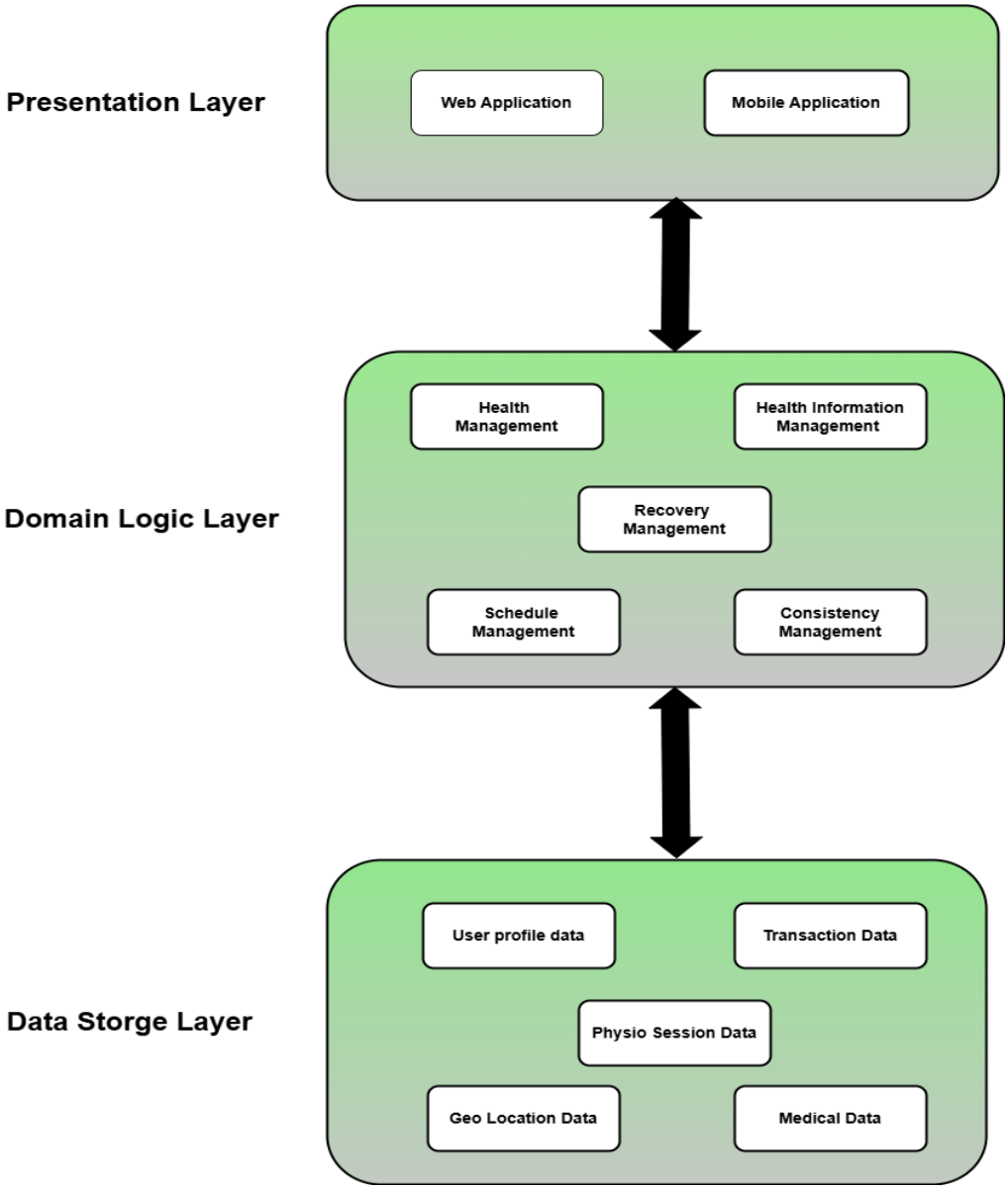


Figure 14: System Architecture Design

6.3 System Design

6.3.1 Class Diagram/Data Flow Diagram (DFD)

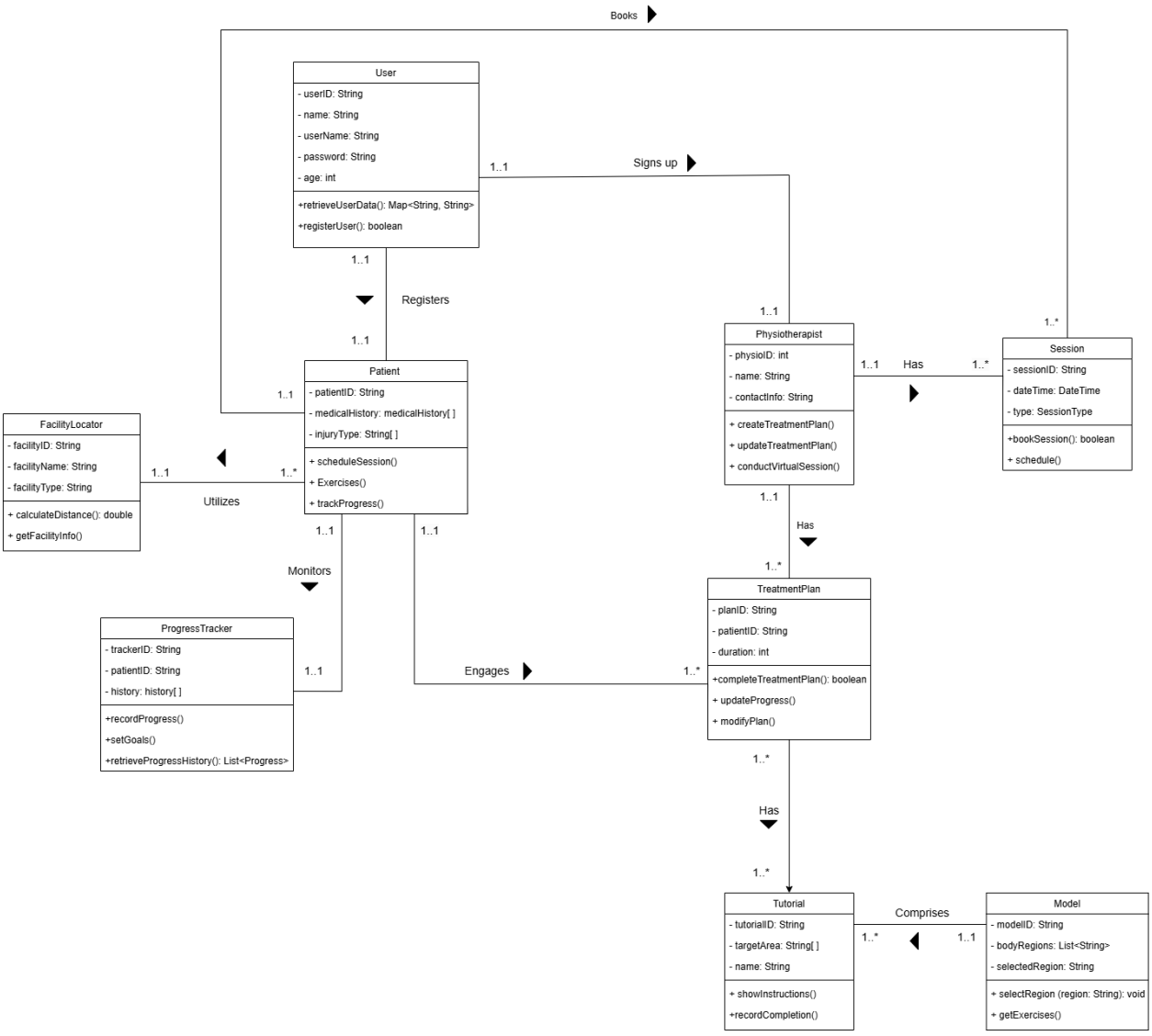


Figure 15::Class Diagram

6.3.2 Sequence Diagram

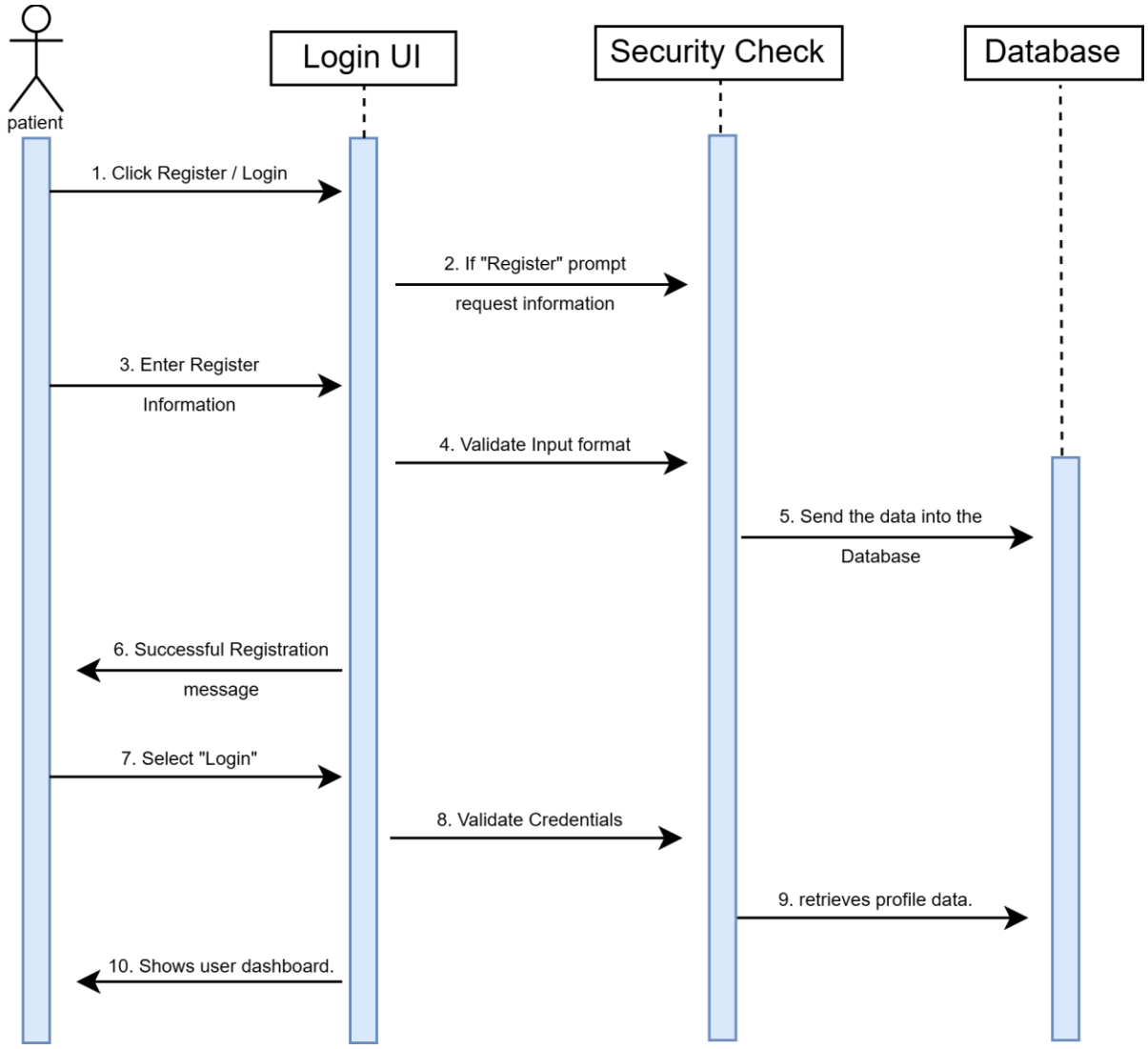


Figure 16 : Sequence Diagram - User Registration and Login

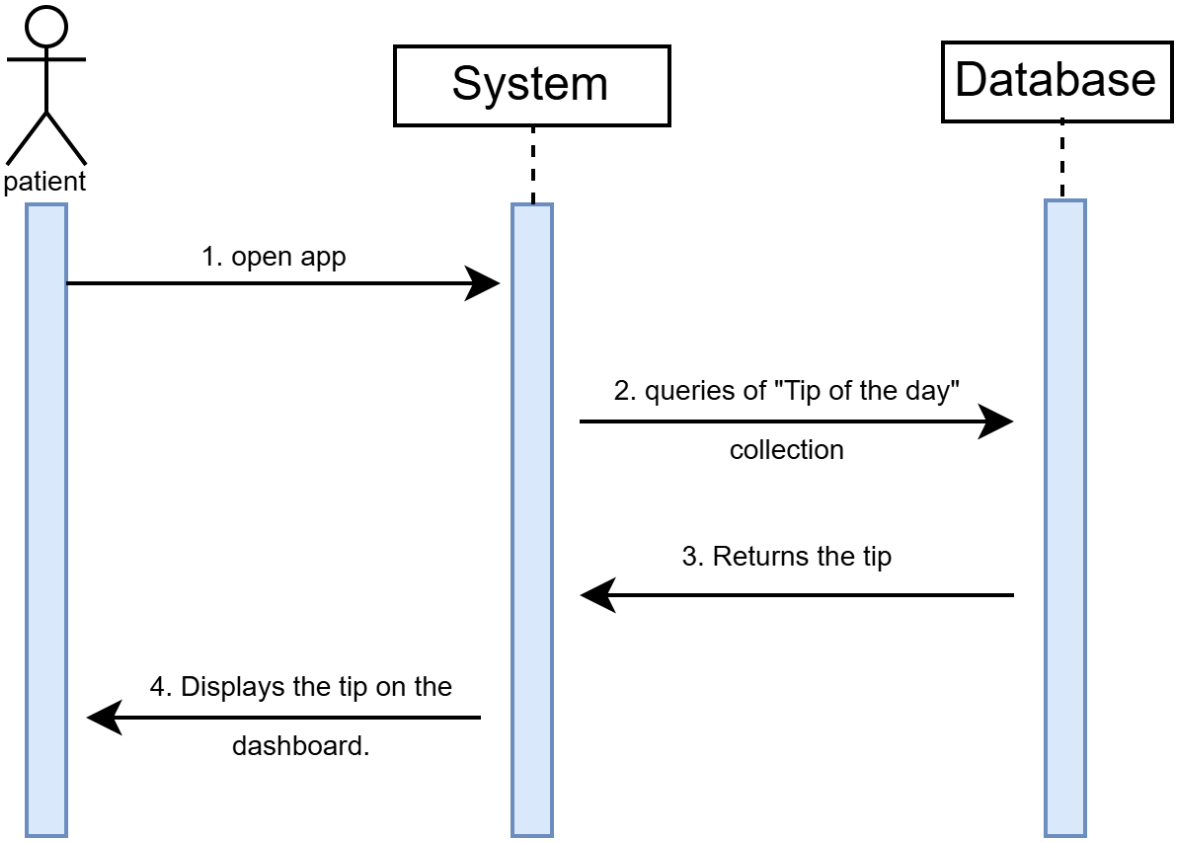


Figure 17 : Sequence Diagram - Tip of the day

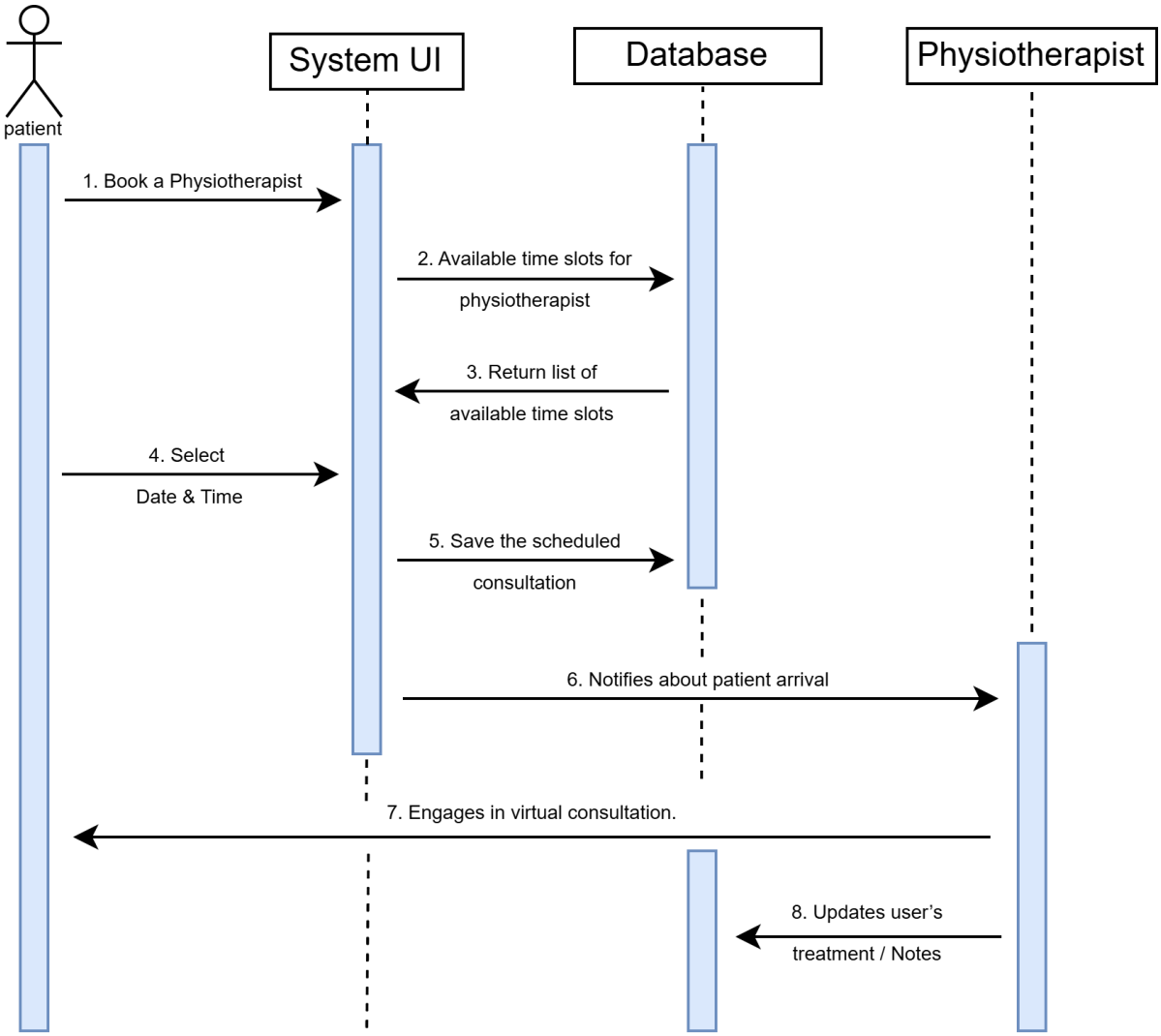


Figure 18: Sequence Diagram - Virtual Consultation

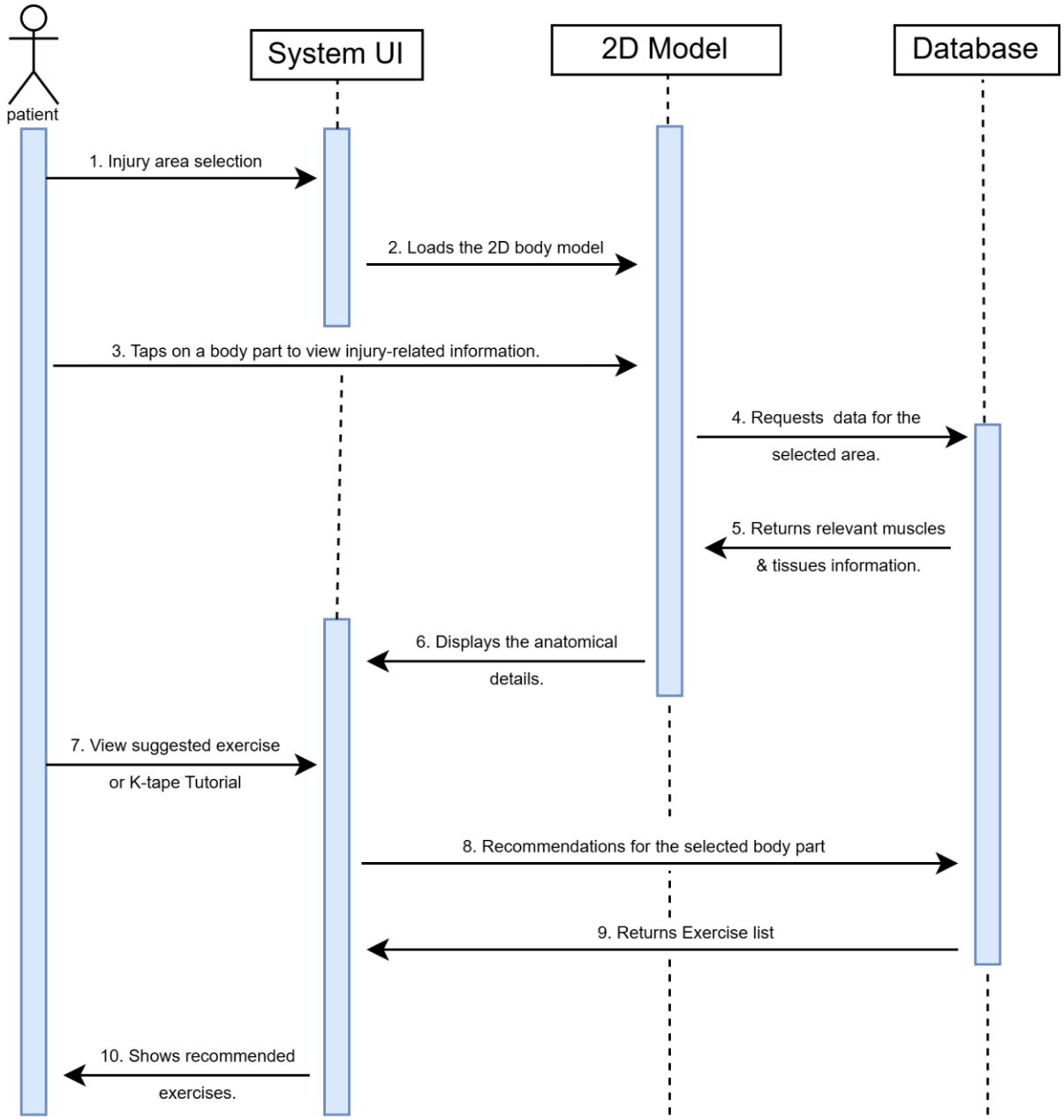


Figure 19 : Sequence Diagram - Injury Area Selection and 2D Model Interaction

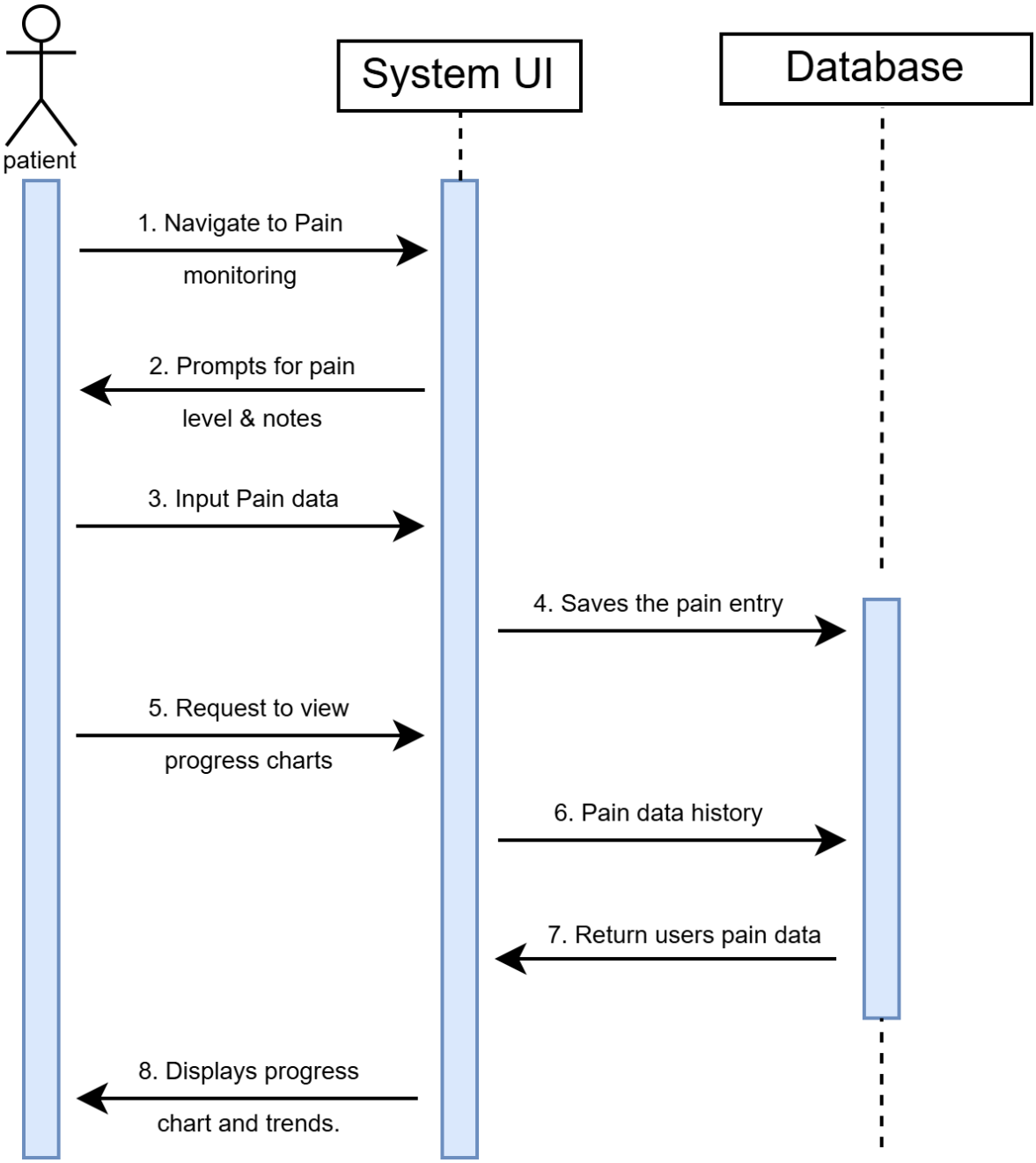


Figure 21 : Sequence Diagram - Progress Tracking

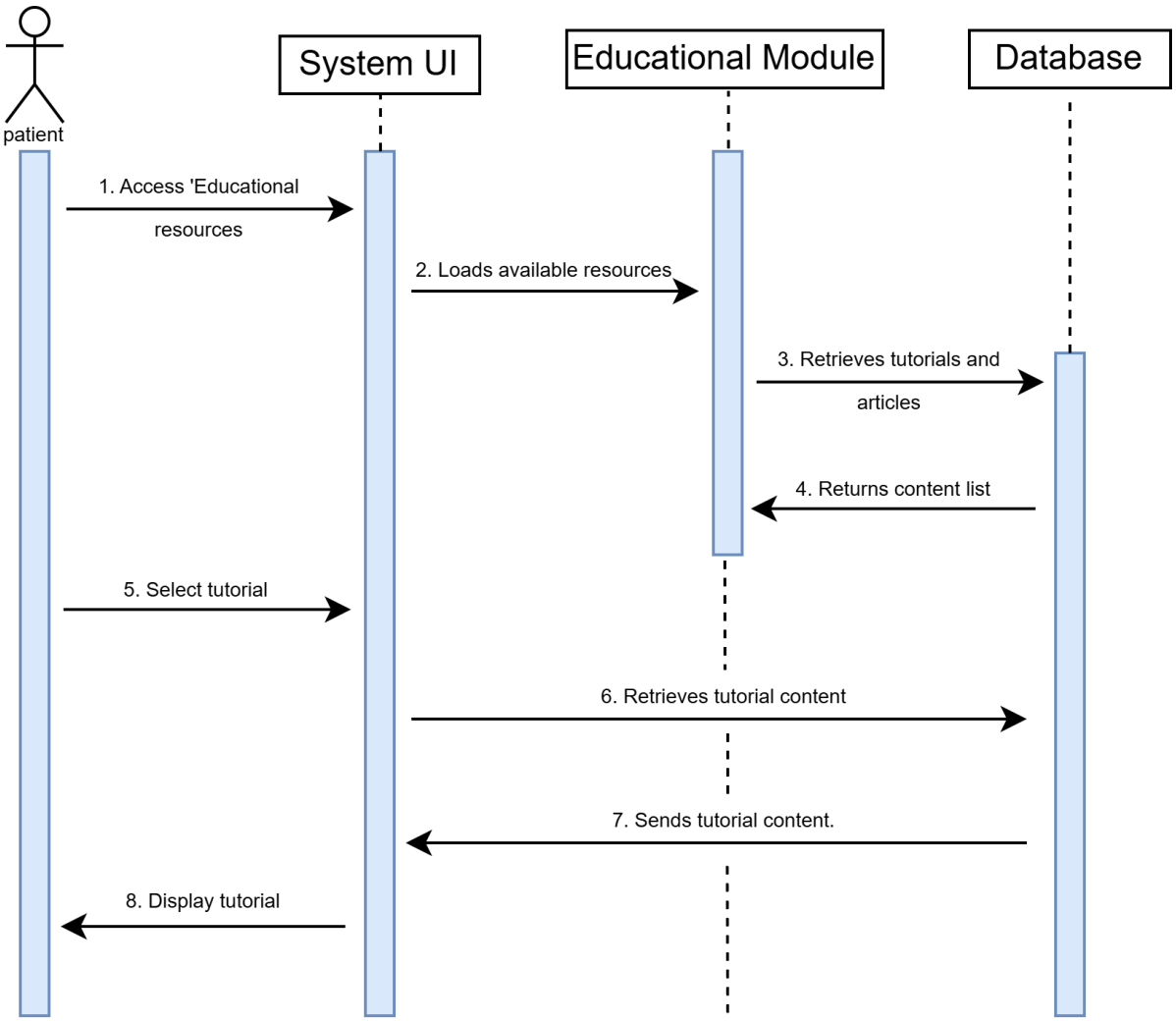


Figure 22 : Sequence Diagram - Educational Resources

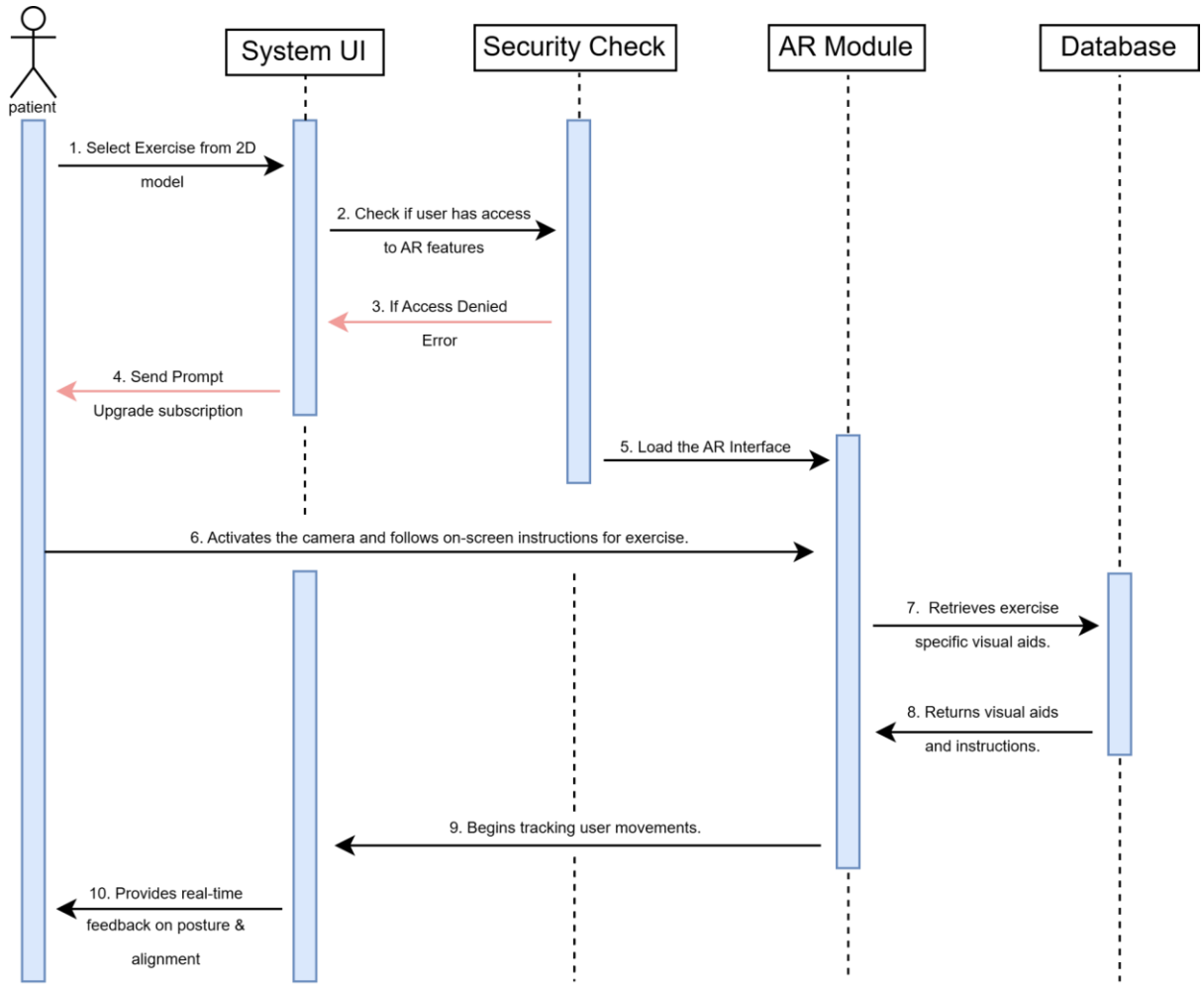


Figure 23 : Sequence Diagram - AR Guidance

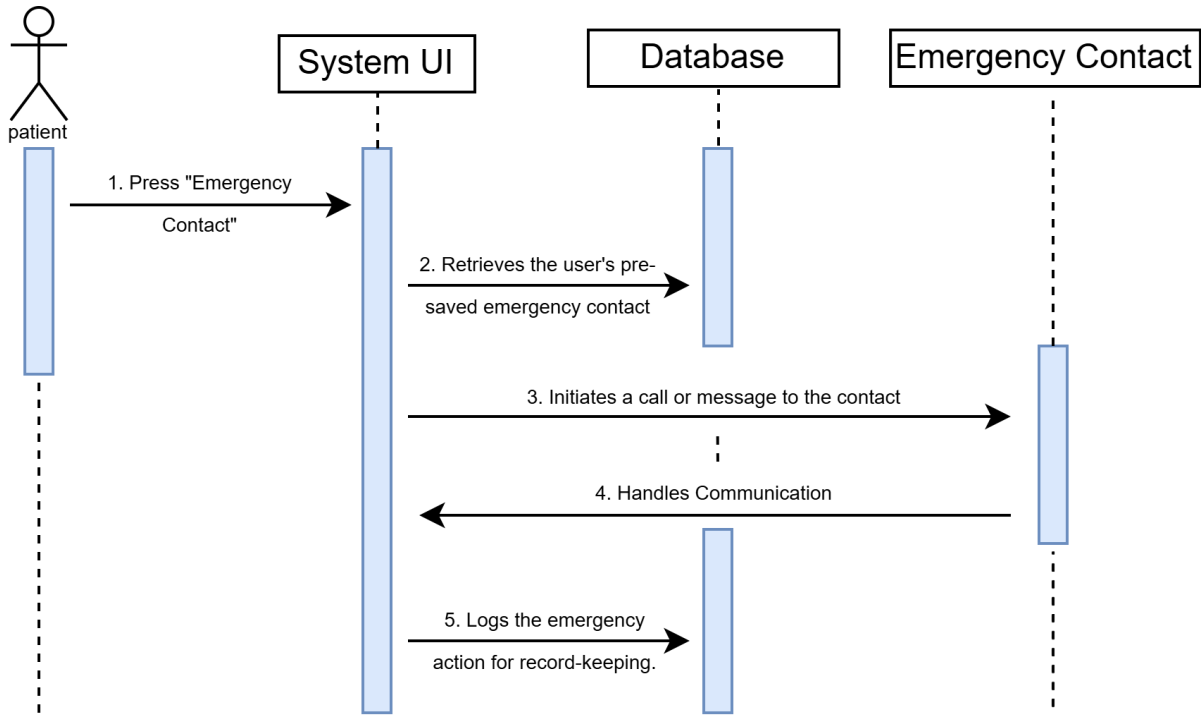


Figure 24 : Sequence Diagram - Emergency Contact

6.3.3 UI Design and mockups



Figure 25:UI Design and mockups -startup page

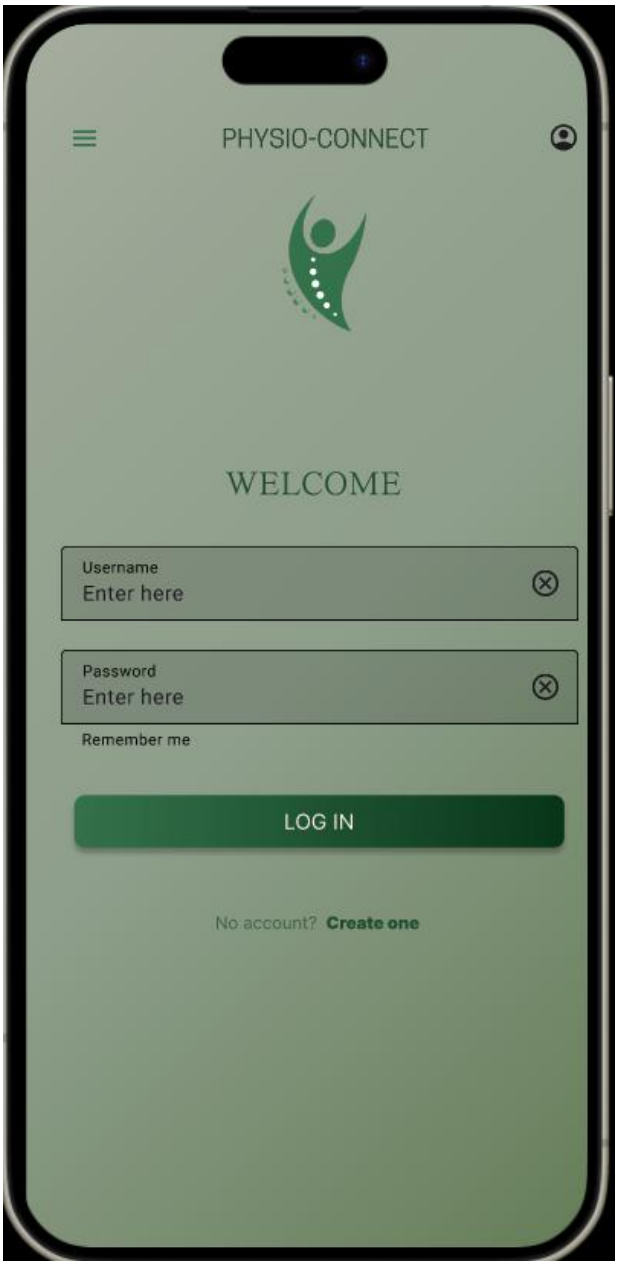


Figure 26:UI Design and mockups - login page



Figure 27:UI Design and mockups -Sign in page

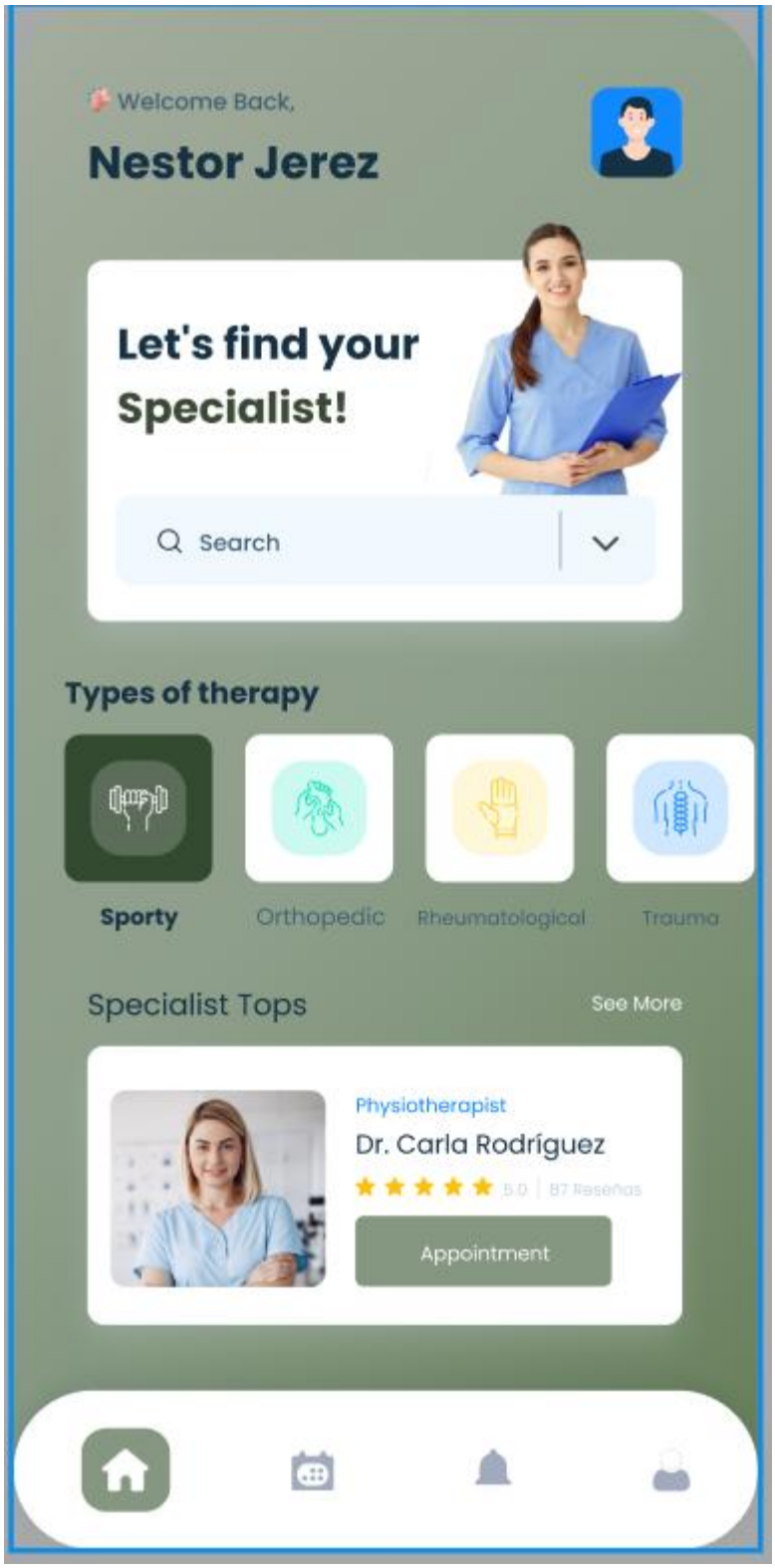


Figure 28 :UI Design and mockups -Home page

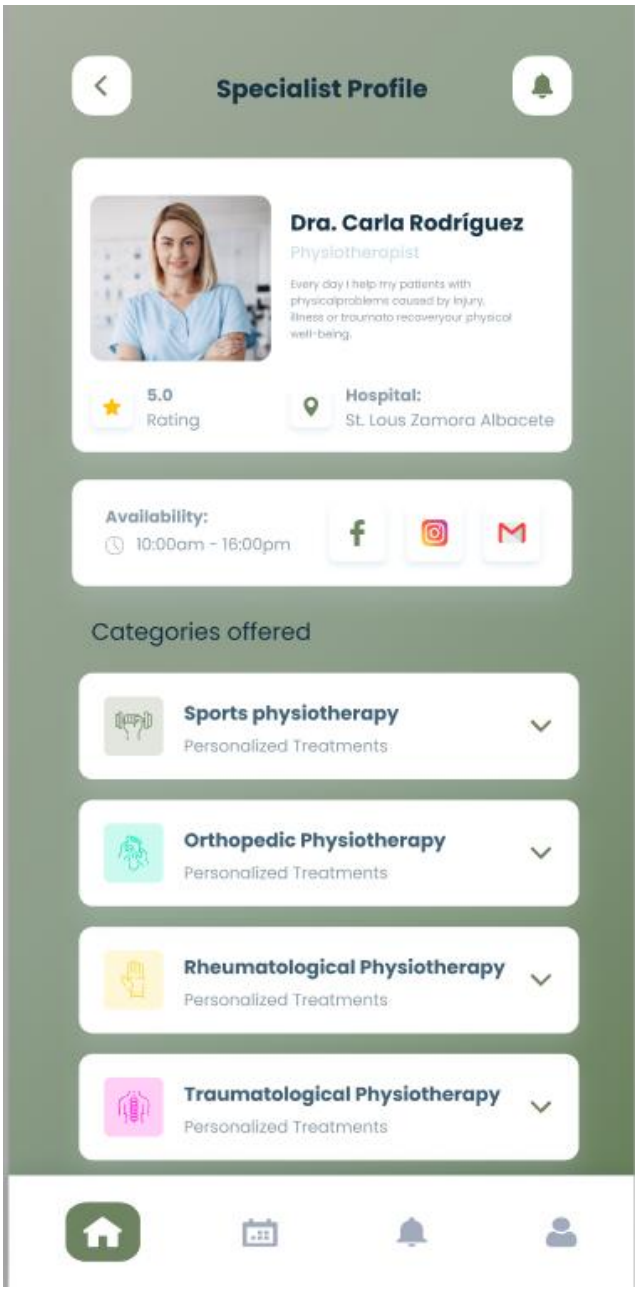


Figure 29:UI Design and mockups -consultation page

6.3.4 Activity Diagram

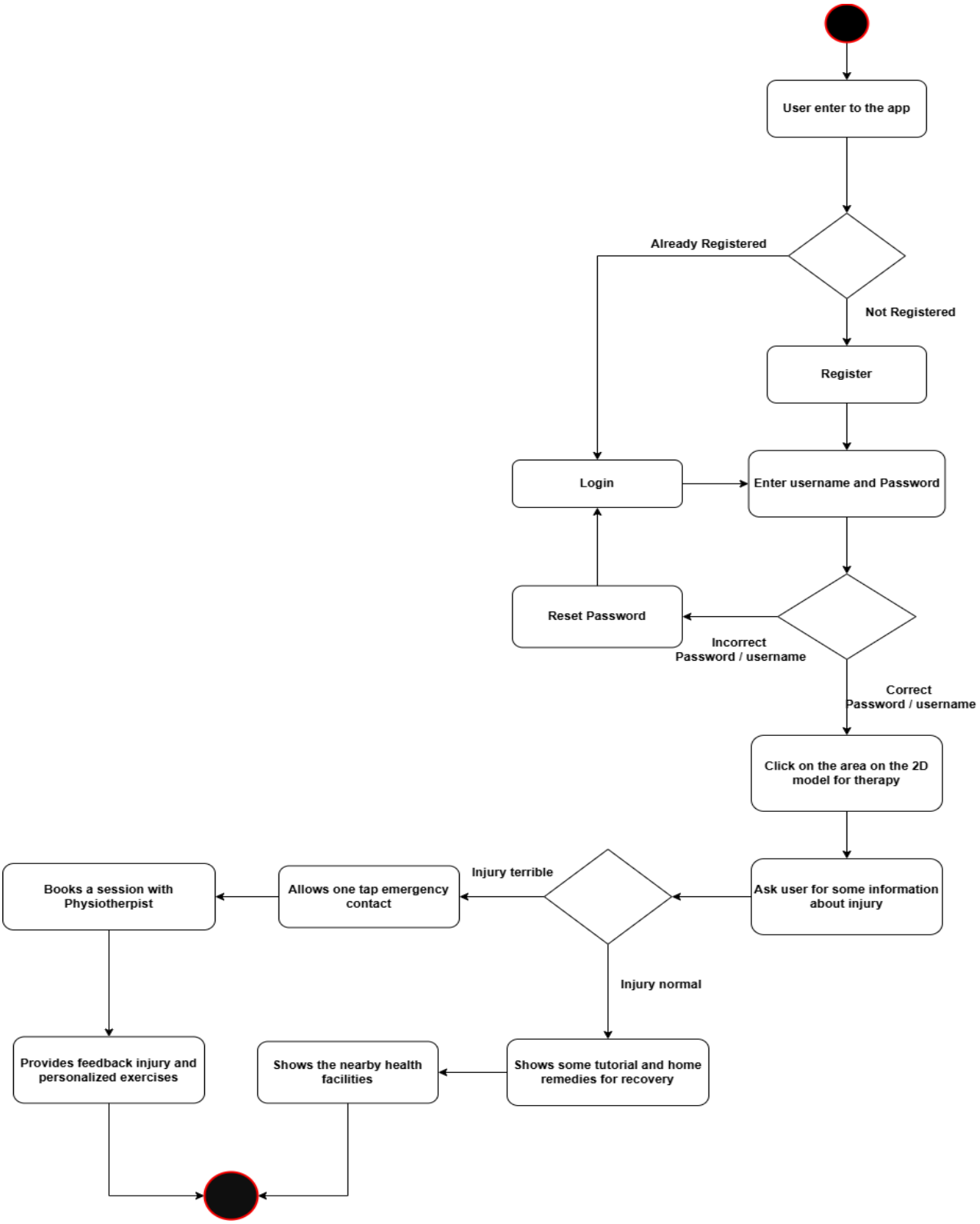


Figure 30 :Activity Diagram

6.4 Chapter Summary

This chapter explores the design and architecture of software systems, starting with an overview of the system architecture, which outlines the organization and interaction of components, it then delves into the key modeling techniques, including class diagrams and data flow diagrams to depict system structure and data flow, and sequence diagrams to visualize dynamic interactions and message flows.

Additionally, the chapter covers UI design and mockups, presenting user interface layouts, and activity diagram, which illustrate workflows and decision points, together, these tools provide a comprehensive framework for designing, visualizing and documenting software systems

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Appendix

What is your gender? *

- ○ Male
- ○ Female

What is your age group? *

- ○ Under 25
- ○ 25-34
- ○ 35-44
- ○ 45-54
- ○ 55 and above

In which profession are your currently employed? *

Your answer

Does your profession involve physical activities or extended hours of sitting or standing? *

- [ ] Physical activities
- [ ] Extended hours of sitting
- [ ] Extended hours of standing
- [ ] All
- [ ] None of the above

How many hours per day do you typically sit or stand while doing work?

- ○ less than 4 hours
- ○ 4-6 hours
- ○ 6-8 hours
- ○ More than 8 hours

Have you ever faced any pains or injuries during your work? *

- ○ Yes
- ○ No

How familiar are you with K taping and similar techniques for injury prevention and recovery? *

1	2	3	4	5	6	7	8	9	10
○	○	○	○	○	○	○	○	○	○

Will you be interested in learning basics physiotherapy techniques to help relieve pain at home with the help of an app? *

- ○ Yes
- ○ No
- ○ Maybe



PhysioConnect

azxxzurrahmaan@gmail.com Switch accounts

* Indicates required question

Email *

Your email address

What features would you find most useful in such app?

Your answer

Would you be interested in setting personal recovery or fitness goals within the app?

- ◯ Yes
- ◯ No

Submit Clear form